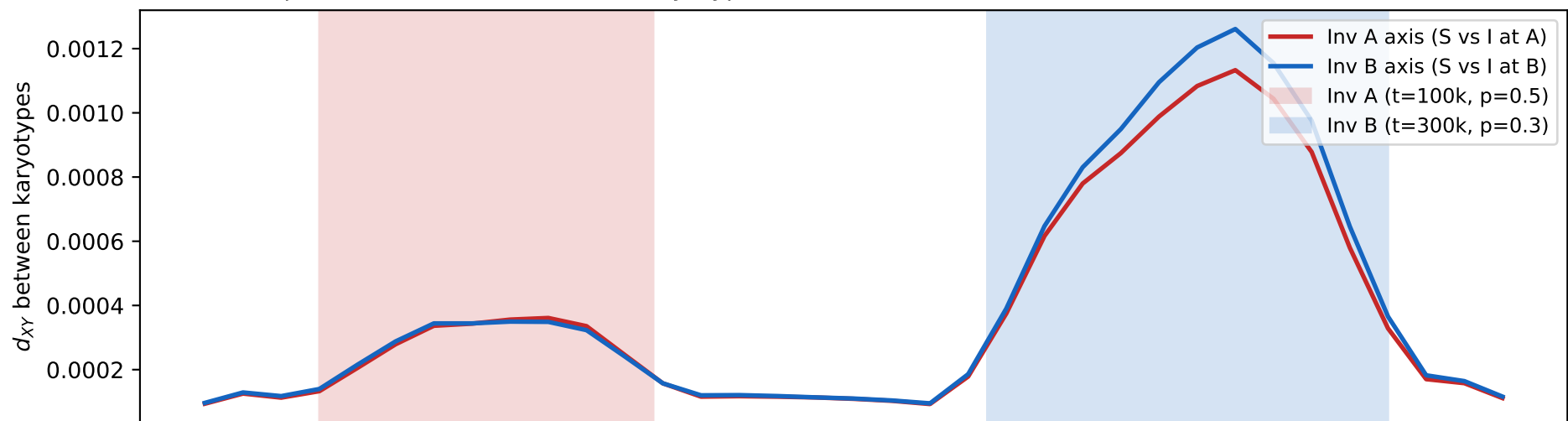


A. Two independent inversions — cross-karyotype d_{XY}



B. Hudson F_{ST} — each inversion elevates its own axis

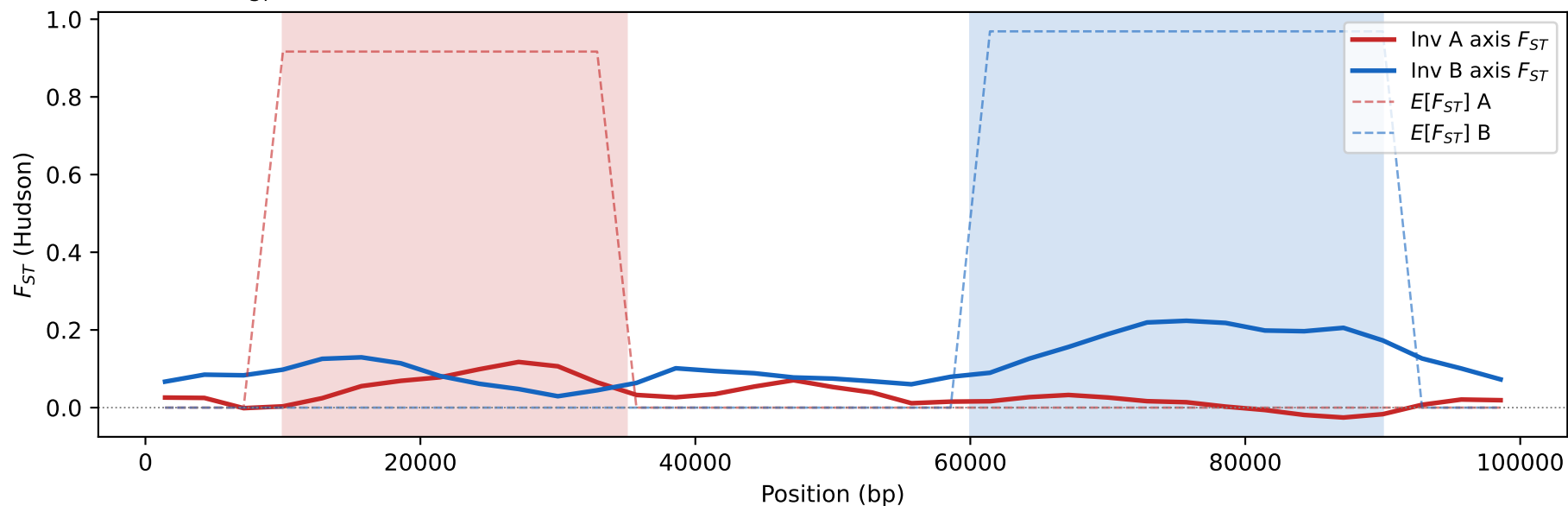


Figure 4. Two independent inversions on the same chromosome, each generating its own cross-karyotype divergence barrier. Inv A (10–35 kb; $t_{inv}=100,000$ gen, $p_{inv}=0.5$, red shading) and Inv B (60–90 kb; $t_{inv}=300,000$ gen, $p_{inv}=0.3$, blue shading). (A) d_{XY} elevated along each S-vs-I axis inside its respective inversion. (B) Hudson F_{ST} shows the same pattern as relative differentiation. Inv B is older and produces a stronger signal. Sample configuration: 5 SS + 5 II + 4 SI + 4 IS (18 haplotypes). Parameters: $N_e=10,000$, $L=100$ kb, $\mu=1e-8$, $r=1e-8$, $\gamma=1e-9$, 80 replicates. Command: `pixi run -e all python examples/make_figures.py`